



**Verified Carbon
Standard**

IKAPA SOLAR PV AND BESS 240 MWP PROJECT



BRUNDTLAND

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Prepared by	<i>Brundtland (Pty) Ltd.</i>

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Project description and location:

The iKapa Solar PV Project (“the Project”) is developed by iKapa Energy Holdings (Pty) Ltd (“the Project Proponent”) and involves the development and operation of a 240 MWp solar photovoltaic (PV) plant combined with a battery energy storage system (BESS) in Saldanha Bay, Western Cape, approximately 140 km from Cape Town. The site is situated within an industrial area and the land secured spans approximately 330 hectares. It is implemented at a single site with multiple integrated technologies, namely large-scale solar PV system and the use of BESS technology to dispatch the renewable energy, ensuring stable and on-demand electricity supply to the grid. The plant will generate and store renewable solar energy, sending energy through South Africa’s carbon-intensive grid to the end users.

This configuration also provides operational flexibility and is particularly advantageous during loadshedding events, as it supports continued electricity delivery and reduces reliance on diesel generators. This is made possible through the BESS that stores generated solar energy and discharges the energy when generation stops. The PV and BESS facility is expected to produce approximately 445 738 MWh of clean electricity annually.

Generation of greenhouse gas emission reductions:

The Project generates renewable electricity by converting solar energy into electrical energy through photovoltaic technology. This is then supplied to BESS technology which stores and dispatches the clean energy to the national grid, thereby displacing electricity that would otherwise be produced using fossil fuel-based generation. By reducing the reliance on coal-based electricity, the Project significantly lowers associated greenhouse gas emissions.

Baseline scenario:

In the absence of the project activity, the equivalent electricity demand would have been met by South Africa’s national grid, which is predominantly powered by fossil fuel-based generation. The grid is largely dependent on coal-fired power stations operated by Eskom, resulting in a high carbon intensive grid electricity.

Estimated emission reductions:

The project activity is expected to generate substantial greenhouse gas emission reductions over its 21-year crediting period. Based on an expected annual renewable energy generation of 445 738 MWh, the Project anticipates an average annual reduction of approximately 341,379 tCO₂e and total reductions of approximately 2,731,030 tCO₂e over the first 7-year crediting period.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/verification	N/A	VCS	N/A	N/A

1.3 Sectoral Scope and Project Type

Sectoral scope ¹	1. Energy Industries (renewable/non-renewable)
Project activity type	Renewable Energy Project

1.4 Project Eligibility

1.4.1 General eligibility

Scope and exclusions under the VCS:

The project site, located in South Africa, produces renewable energy thereby reducing energy demand on the carbon-intensive national grid. By displacing grid electricity with renewable energy, the Project contributes to a decrease in CO₂ emissions - one of the seven greenhouse gases identified under the Kyoto Protocol.

As such, the Project falls within the scope of the VCS Programme and is therefore eligible under the VCS as it meets the criteria outlined in paragraph 2.1.1 of the VCS Standard, Version 4.7. Additionally, this Project does not fall within any of the excluded project categories listed in Table 1 as shown below:

Table 1 Excluded Project Activities under the VCS Standard (V4.7)

Activity	Applicability
Grid-connected electricity generation using hydro power plants	N/A. The project activity is not electricity generation using hydro power plants.
Grid-connected electricity generation using wind, geothermal, or solar PV power plants.	This is a grid-connected electricity generation activity using solar PV technology to supply BESS technology, which then stores and dispatches the

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

	clean energy to the grid. In accordance with the VCS Standard, the Project is eligible under the scope of grid-connected renewable energy activities and does not fall under the excluded category, as the exclusion does not apply where such activities utilise energy storage systems, such as BESS.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	N/A. The Project and project activity produce electricity through renewable solar PV and BESS technology and thus does not use recovered waste heat for any purposes.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue, or forest residue biomass.	N/A. The Project and project activity produce electricity through renewable solar PV and BESS technology and thus does not use biomass.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	N/A. The Project and project activity produce electricity through renewable solar PV and BESS technology and thus does use fossil fuels or activities that involve switching from a higher to a lower carbon content fossil fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	N/A. The Project and project activity produce electricity through renewable solar PV and BESS technology and thus does not involve the replacement of electric lighting with more energy-efficient electric lighting.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	N/A. The Project and project activity produce electricity through renewable solar PV and BESS technology and thus does not involve the installation or replacement of electricity transmission lines and/or energy-efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions.	N/A. The Project and its associated activity instances reduce CO ₂ emissions by generating renewable electricity through solar PV and BESS

technology, thereby displacing emissions that would otherwise result from fossil fuel-based power generation and thus does not involve the reduction of hydrofluorocarbon-23 (HFC-23) emissions.

Methodology eligibility

There are no approved VCS specific methodologies under the VCS which could be applied to this Project, therefore CDM was used to identify the applicable methodology. ACM0002 '*Grid-connected electricity generation from renewable sources*' methodology (Version 22.0), which has been approved and is active under the VCS Programme, is the most applicable methodology for this project activity. See Section 3.2 below for further details.

The Project meets the applicability conditions of ACM0002, which is a large-scale methodology with no installed capacity limit. This is a grid-connected electricity generation activity using solar PV technology to supply the BESS and is therefore eligible and not excluded. The Project consists of a single renewable electricity generation facility at one location and does not form part of a grouped project or a broader activity.

Compliance with Deadlines

The Project shall meet the requirements of the VCS, relating to the pipeline listing deadline, the opening meeting with the VVB and the validation deadline under the guidance of the VCS Standard.

1.4.2 AFOLU project eligibility

This section does not apply, as the Project is not classified as an AFOLU activity.

1.4.3 Transfer project eligibility

Not applicable – this Project is not a transfer project.

1.5 Project Design

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

This is not a grouped project.

1.6 Project Proponent

Organization name	iKapa Energy Holdings (Pty) Ltd.
Contact person	Mr. Robert Longhi
Title	Finance Director (Non-Executive)
Address	13 Clingendael Close, 7945 Tokai, Cape Town
Telephone	+27 21 713 0477
Email	rlonghi81@gmail.com

Organization name	Brundtland (Pty) Ltd.
Contact person	Mr. Henk Sa
Title	Director
Address	Firestation, 16 Baker Street, Rosebank, 2196, Gauteng, South Africa
Telephone	+27 11 447 7892 / +27 78 459 5748
Email	henk.sa@brundtland.co.za

1.7 Other Entities Involved in the Project

There are no other entities involved in the Project.

1.8 Ownership

iKapa Energy Holdings (Pty) Ltd. holds the legal right to manage and operate the activities and facilities associated with the Project. The company holds all necessary statutory, property, or contractual rights to the plant, equipment, and processes responsible for generating GHG emission reductions or removals, thereby confirming iKapa Energy Holdings as the project owner. Additionally, the company holds legal land use rights through land lease agreements valid for the full duration of the project's lifetime.

1.9 Project Start Date

Project start date	01-December-2026
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Justification	The 'start date' of the Project is defined as the first date on which generation of energy occurs following no previous generation from that specific site. The Project Proponent has indicated that construction is expected to end late in 2026 on which generation can commence.
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1.10 Project Crediting Period

Crediting period	☒ <i>Seven years, twice renewable</i> ☐ <i>Ten years, fixed</i> ☐ <i>Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)</i>
Start and end date of first or fixed crediting period	01-Decemebr-2026 to 30-November-2033

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☐ < 300,000 tCO₂e/year (project)
☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-December-2026 to 31-December-2026	32,512
01-January-2027 to 31-December-2027	390,147
01-January-2028 to 31-December-2028	390,147
01-January-2029 to 31-December-2029	390,147
01-January-2030 to 31-December-2030	390,147
01-January-2031 to 31-December-2031	390,147

01-January-2032 to 31-December-2032	390,147
01-January-2033 to 30-November-2033	357,635
Total estimated ERRs during the first or fixed crediting period	2,731,030
Total number of years	7
Average annual ERRs	341,379

1.12 Description of the Project Activity

The project activity is a single-site renewable energy power plant that integrates large-scale solar photovoltaic (PV) technology with a Battery Energy Storage System (BESS) to reliably displace fossil fuel-based electricity on South Africa's national grid. As previously noted, most of the electricity supplied through the grid is generated by the state-owned utility, Eskom, the largest energy producer in the country. Eskom's electricity is primarily derived from coal-fired power stations, one of the most carbon-intensive energy sources, therefore, making it the primary contributor to South Africa's greenhouse gas (GHG) emissions.

With a combined capacity of 240 MWp, the iKapa Saldanha Bay facility is expected to generate approximately 445,738 MWh of clean electricity per year. This renewable generation directly displaces an equivalent amount of electricity from South Africa's carbon-intensive grid. Consequently, the Project is projected to reduce national carbon emissions by approximately 341,379 tonnes of CO₂ equivalent (tCO₂e) on average each year.

Solar PV system: The solar PV system consists of LONGi-supplied photovoltaic panels and Huawei inverters. The panels are composed of photovoltaic (PV) cells that absorb sunlight and convert it into direct current (DC) electricity. The DC electricity is then passed through an inverter, which converts it into low voltage alternating current (AC) electricity suitable for transmission and distribution via the grid.

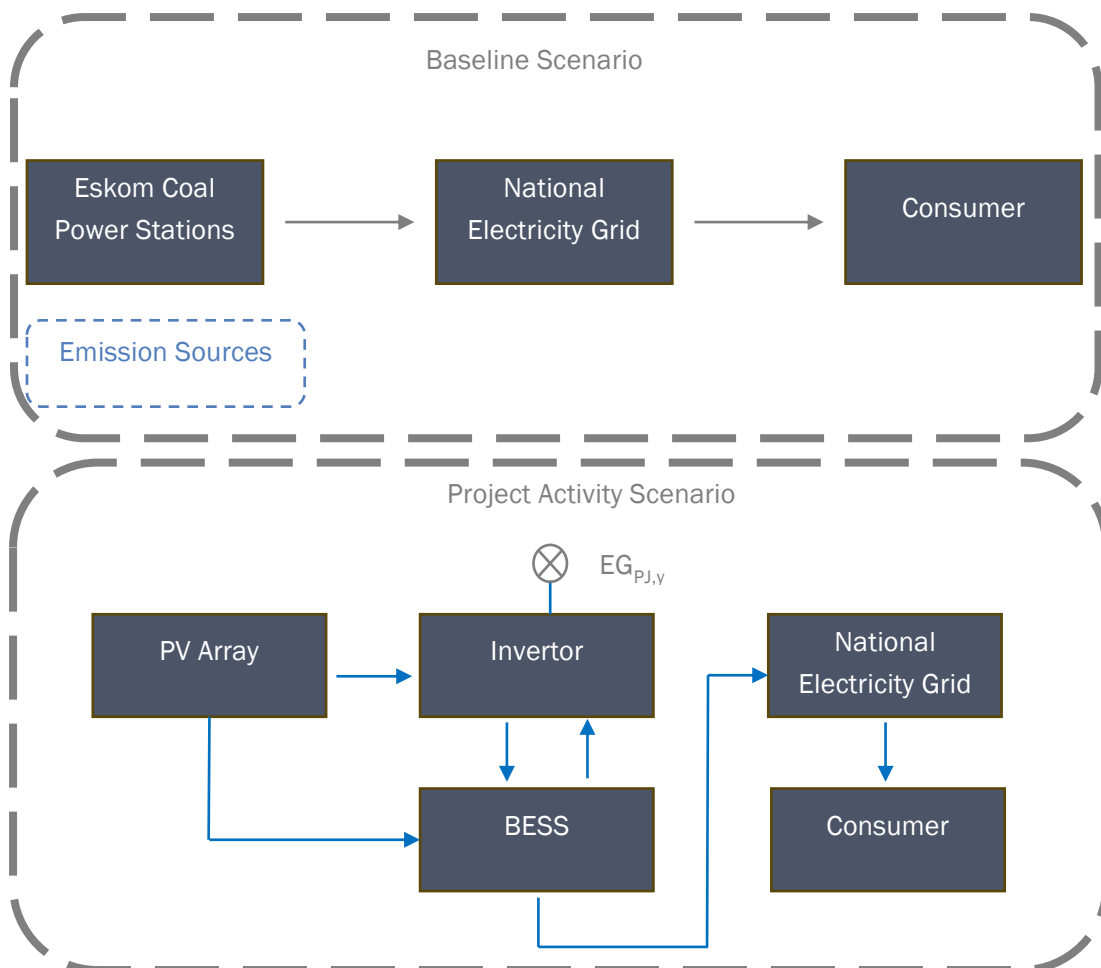
Solar energy generation is variable and depends on environmental conditions such as sunlight availability, cloud cover, haze, and dust. Furthermore, solar generation does not always align with electricity demand, leading to curtailment (where excess energy is deliberately reduced or wasted to protect grid stability) during periods of oversupply, and shortfalls during peak demand or nighttime.

BESS technology integration: To address these intermittent supply/demand challenges, the iKapa Project integrates a BESS, typically based on lithium-ion technology due to its high energy density, rapid response capability, and commercial viability. The BESS includes a Battery Management System (BMS) that optimizes charging and discharging cycles, monitors system performance, and ensures operational safety and efficiency.

During periods of high solar output and low grid demand, excess DC electricity from the solar panels is diverted to the BESS for storage. The stored energy can then be discharged during periods of low solar generation, such as at night, during cloudy weather, or during grid outages (e.g., load shedding), thereby improving power reliability. When electricity is discharged from the BESS, it follows a similar pathway as real-time generation: it is first converted from DC to AC via an inverter before feeding into the national grid..

The solar PV panels have a design lifetime of 25 to 30 years, and the Huawei inverters are expected to operate for approximately 15 years with routine maintenance. The lithium-ion battery systems have an expected lifetime of 10 to 15 years, depending on cycling patterns. The inverters operate at approximately 97% efficiency, and the BESS has a round-trip efficiency of 85–90%. The expected capacity factor of the PV system is approximately 21%, based on site-specific solar resource assessments.

The Project is expected to begin construction in 2026, with commercial operations commencing thereafter. By integrating utility-scale solar PV with advanced battery storage, the Project offers a flexible, resilient, and low-carbon solution to South Africa’s energy needs. It not only contributes to national emissions reductions but also enhances energy security and grid stability in a context where coal dependency and load shedding remain ongoing challenges.



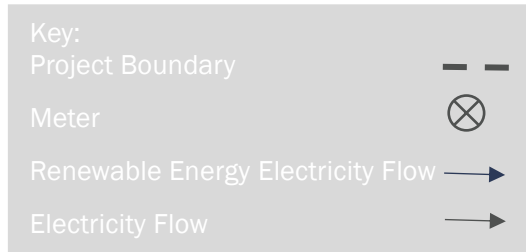


Figure 1: Equipment arrangement and energy flows

1.13 Project Location

The project is in Saldanha Bay in the Western Cape as indicated previously, with the coordinates 32°56'39"S latitude and 18°02'26"E longitude. The project will be implemented on leased land spanning 330 hectares, in the region's industrial zone.

KML file of the project site will be provided on the VERRA Registry.

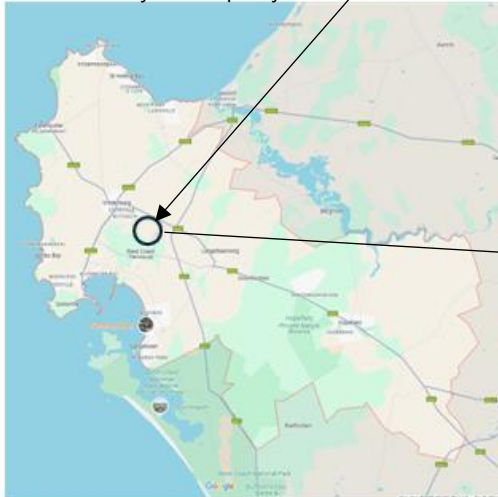
South Africa



Western Cape Province



Saldanha Bay Municipality



iKapa project site



Figure 2: Project site location – South Africa, Western Cape, Saldanha Bay Municipality

1.14 Conditions Prior to Project Initiation

The solar PV and BESS project has been launched with the primary aim of strengthening electricity supply for consumers, especially considering South Africa's enduring energy security challenges. Since 2007, the country has grappled with frequent load shedding, resulting in inconsistent and unreliable power availability. Escalating electricity tariffs have further intensified the demand for affordable, dependable and sustainable energy alternatives.

The conditions prior to the project initiation correspond to the project baseline scenario which is further described in section 3.4.

This project is exclusively focused on generating clean, renewable energy and supporting climate change mitigation efforts by replacing fossil fuel-derived electricity from the national Eskom grid, which remains predominantly dependent on non-renewable sources.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

No restrictions under South African laws or regulatory frameworks have been identified that would impede the Project's development.

National Environmental Management Act (No. 107 of 1998)	The Project complies with the National Environmental Management Act (No. 107 of 1998) (NEMA), which provides the legislative framework for integrated environmental management in South Africa. To satisfy the requirements of the Act and associated Environmental Impact Assessment (EIA) Regulations, an environmental assessment process will be initiated. This process is established to ensure that all environmental impacts of the Project are fully considered and addressed.
Occupational Health and Safety Act, 1993	The Project fully complies with the Occupational Health and Safety Act, 1993 (Act No. 85 of 1993), including Regulation 7(1) of the Electrical Installations Regulations, 2009, to ensure the well-being of personnel in connection with electrical installations and the conduct of related installation operations.

Electricity Regulation Act (No. 4 of 2006) and NERSA Licensing	The Project complies with the Electricity Regulation Act (No. 4 of 2006), which governs the licensing of electricity generation activities in South Africa. With an installed capacity exceeding 100 MW, the Project falls within the scope of activities requiring a generation license from the National Energy Regulator of South Africa (NERSA). A full license application will be submitted in line with NERSA's requirements, supported by all necessary technical, financial, and environmental documentation to ensure full regulatory compliance.
National Energy Act (No. 34 of 2008):	The Project supports the objectives of the National Energy Act (No. 34 of 2008) by contributing to energy security, promoting cleaner electricity generation, and enabling greater diversity in South Africa's energy mix. Through its use of solar PV and BESS, the Project enhances grid stability and aligns with national energy planning principles focused on sustainability, efficiency, and long-term resilience.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.1 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

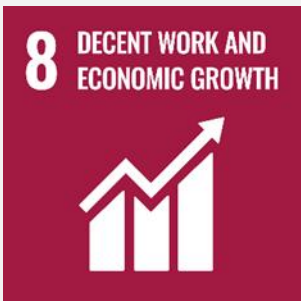
If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

1.18 Sustainable Development Contributions

SDG 13		
	Target 13	Take urgent action to combat climate change and its effects.
	Indicator	Tonnes of avoided greenhouse gas emissions (tCO ₂ e)
The Project will displace fossil-fuel based electricity generation with clean, renewable energy through solar. South Africa, as a signatory to the Paris Agreement, has committed to achieving Net-Zero emissions by 2050. As part of this commitment, South Africa has developed and submitted its first Nationally Determined Contribution (NDC), which identifies the growth of the renewable energy sector as a national priority. Through its large-scale renewable energy power plant, iKapa contributes considerably to South Africa's national climate objectives and the global goal of Climate Action by displacing almost 412,803 tCO₂e annually. Monitoring of this contribution will be carried out by the iKapa Saldanha Bay team, by tracking the total renewable energy generated (in MWh). This data will be used to calculate the corresponding emissions avoided (in tonnes of CO₂e) and will be included in the Project's monitoring reports.		

SDG 8		
	Target 8.3	Promote policies to support job creation and growing enterprises
	Indicator	ZAR spent on skilled labourers or number of employees that are employed for commissioning and maintenance of solar PV facilities.

The Project requires the construction, operation and maintenance of solar PV system and BESS integration that will generate job opportunities. These roles often require specialised skills, which promotes the development of a more qualified workforce and encourages individuals to pursue training (e.g.: to become solar technicians). By both increasing job opportunities and improving job quality, the Project contributes to broader economic development in South Africa.

Job creation is a national priority for the government of South Africa, given the high unemployment rate. Government initiatives such as the National Development Plan, the Labour Activation Programme, and targeted investment in infrastructure and public works all seek to address this challenge and stimulate the economy. In this context, iKapa's Project aligns well with the country's objectives for job creation, decent work and economic growth.

Monitoring of SDG 8 contributions will need to be undertaken by iKapa and be included as part of the regular monitoring reports. Key indicators should include number of employees hired and any additional financial contributions toward training and upskilling workers.

SDG 7



Target 7.2	Increase Global Percentage of Renewable Energy
Indicator	Renewable electricity generated (MWh)

The Project will displace emission intensive grid electricity through the generation of zero-emission solar PV electricity. This will contribute towards improving access to a reliable, clean source of energy in South Africa. Moreover, iKapa forms part of a shifting energy landscape in South Africa where, more private power producers and other key players (e.g.: energy aggregators) are allowed to participate. Through a more liberalised market, electricity prices are kept affordable due to competition in the market benefitting South African consumers.

The country's Integrated Resource Plan (IRP) outlines the national strategy for meeting future energy demand, with a growing emphasis on renewable energy sources. iKapa's work supports this national agenda by increasing the supply of clean, reliable electricity.

Monitoring of SDG 7 will be managed by iKapa, using the same MWh-based tracking of renewable energy generation outlined in the PD.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

To be completed in full validation version

1.19.2 Commercially Sensitive Information

To be completed in full validation version

1.19.3 Further Information

To be completed in full validation version

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

To be completed in full validation version

2.1.2 Stakeholder Consultation and Ongoing Communication

To be completed in full validation version

2.1.3 Free Prior and Informed Consent

To be completed in full validation version

2.1.4 Grievance Redress Procedure

To be completed in full validation version

2.1.5 Public Comments

To be completed in full validation version

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

To be completed in full validation version

2.2.2 Risk Assessment

To be completed in full validation version

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

To be completed in full validation version

2.3.2 Human Rights

To be completed in full validation version

2.3.3 Indigenous Peoples and Cultural Heritage

To be completed in full validation version

2.3.4 Property Rights

To be completed in full validation version

2.3.5 Benefit Sharing

To be completed in full validation version

2.4 Ecosystem Health

To be completed in full validation version

2.4.1 Rare, Threatened, and Endangered Species

To be completed in full validation version

2.4.2 Introduction of Species

To be completed in full validation version

2.4.3 Ecosystem Conversion

To be completed in full validation version

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table below includes the methodology(s), tools and modules applied to the project activity.

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	CDM TOOL 01	Tool for the demonstration and assessment of additionality	7.0
Tool	CDM TOOL 07	Tool to determine the emission factor for an electricity system	7.0
Tool	VT0011	Electricity System Emission Factors	1.0
Tool	CDM TOOL 27	Investment Analysis	14.0

3.2 Applicability of Methodology

The project activity(s) meets each of the applicability conditions of the methodology(s), tools and modules applied by the project (where applicable) as described below:

Methodology ID	Applicability condition	Justification of compliance
ACM0002	Paragraph 5: This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s); or	The project activity involves construction and operation of a new solar power plant integrated with BESS, at a site that had no renewable energy prior to the implementation of the project. Condition (a) would be most applicable to the project as the iKapa Saldanha Bay power plant is a “greenfield power plant”. Therefore, the project activity meets this applicability condition of the methodology.

	(f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	
ACM0002	Paragraph 7: In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic¹ or wind power plant(s)/unit(s) (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s) (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s) (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	The project activity includes the integration of BESS to the Solar PV system at a site that had no renewable energy prior to the implementation of the Project. Condition (a) therefore would be the most applicable as the Saldanha Bay power plant integrates BESS with a greenfield power plant. Therefore, the project activity meets the applicability condition of the methodology.
ACM0002	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power	This project activity involves the construction and operation of a solar power plant/unit. For that reason, Condition (a) is relevant for

	plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit	the Project and therefore meets this application criteria of the methodology.
ACM0002	Paragraph 8: The methodology is applicable under the following conditions: (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit must have started commercial operation prior to the start of a minimum historical reference period of five years. The reference period is used for the calculation of baseline emissions and defined in the baseline emission section. Furthermore, no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity	Not applicable as the Project is on a site that had no renewable energy prior to the implementation of the Project therefore not a capacity addition, retrofit, rehabilitation or a replacement.
ACM0002	Paragraph 8: The methodology is applicable under the following conditions: (c) In case of Greenfield project activities applicable under paragraph 7(a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g., by referring to feasibility studies or investment decision documents)	This Project involves the integration of BESS to the solar PV system. BESS integration ensures the required supply of electricity under any power purchase agreement considering the intermittency of solar renewable energy. Therefore, BESS is an integral part of the project design. Therefore Condition (c) is relevant for this Project and thus, this applicability methodology is met
ACM0002	Paragraph 8: The methodology is applicable under the following conditions:	Excess electricity generated by the solar PV system is used to charge the BESS component on site . Therefore, Condition (d) is relevant

	(d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g., week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period	for the Project and this applicability condition is met as the BESS is charged by renewable energy as required.
ACM0002	Paragraph 8: The methodology is applicable under the following conditions: (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	This is not applicable as the Project does not involve a PSP as part of its project design and therefore, Condition (e) is not relevant for the Project.
ACM0002	Paragraph 9: In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs	Not applicable as the Project does not involve a hydro power plant as part of its project design.

	(b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m² (c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m² (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², and all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² are: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
ACM0002	Paragraph 10: In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project	Not applicable as the Project does not involve a hydro power plant as part of its project design.

	(b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
ACM0002	Paragraph 11: In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	This is not applicable as the Project does not involve a PSP as part of its project design
ACM0002	Paragraph 12: The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	This is not applicable as the Project does not involve a switch from fossil fuels to renewable energy at the site of the project activity nor the operation of a biomass fired power plant/units.
ACM0002	Paragraph 13: In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology	Not applicable as the Project is on a site that had no renewable energy prior to the implementation of the Project therefore not a capacity

	is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”	addition, retrofit, rehabilitation or a replacement.
ACM0002	In addition, the applicability conditions included in the tools referred to below apply.	The Project meets the applicability conditions of the tools referenced in the methodology. These have been shown below.

Tool ID	Applicability condition	Justification of compliance
CDM TOOL 01	Paragraph 9: The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The Project does not involve the proposal of a new methodology as ACM0002, the applied methodology is an existing methodology. Therefore, the Tool is applicable for the Project and iKapa therefore has applied TOOL 01 for the demonstration of additionality as required. The demonstration of the applicability of TOOL 01 along with additionality has been included in the PD.
CDM TOOL 01	Paragraph 10: Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	In line with the ACM0002 methodology section 5.3.2, the demonstration of additionality is required and therefore the Tool is applicable for the Project. The demonstration of additionality has been included in the PD..

Tool ID	Applicability condition	Justification of compliance
CDM TOOL 07	Paragraph 3: This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	The project activity involves the construction and operation of solar PV power plant integrated with BESS. The generated renewable energy (through the BESS) supplies to the grid therefore displacing fossil-fuel based grid electricity. This applicability condition is therefore met. The tool has been used to estimate the OM, BM and CM for the project activity.
CDM TOOL 07	Paragraph 4: Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	The emission factor for the project activity has been determined only for grid power plants and therefore meets the applicability condition. The estimation of the emission factor has been included as part of the emissions reductions calculations made available to the VVB.

CDM TOOL 07	Paragraph 5: In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable as TOOL VT0011 removes this applicability condition. Moreover, South Africa, the host country of the Project, is not an Annex 1 country nor is this project a CDM project.
CDM TOOL 07	Paragraph 6: Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not applicable as the project activity does not include the use of biofuels.

Tool ID	Applicability condition	Justification of compliance
TOOL VT0011	This tool is applied to calculate the CO2 emission factor of an electricity system whose dispatch and generation are affected by a project activity. It is globally applicable.	The Project displaces part of the electricity provided by the grid therefore, affecting overall generation supplied by the grid. This applicability condition is therefore, met and the CO ₂ emission factor has been calculated under the guidance of VT0011 and TOOL 07.
TOOL VT0011	All applicability conditions of the most recent version of TOOL07 must be met, except for Applicability Condition 5, which is removed under this revision	The applicability conditions as outlined by TOOL 07 have been met and shown in the table above.
TOOL VT0011	This tool must be used in conjunction with the most recent version of TOOL07. All the procedures and requirements of TOOL07 must be applied unless this tool indicates otherwise.	TOOL VT0011 has been used in conjunction with TOOL 07 and demonstrated in the PD.

Tool ID	Applicability condition	Justification of compliance
TOOL 27	Paragraph 2: This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the demonstration of additionality	An investment analysis has been chosen and applied for the demonstration of additionality therefore, this applicability condition is met.

	and/or the identification of the baseline scenario.	
TOOL 27	Paragraph 3: In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The requirements for demonstration of additionality provided for in the ACM0002 methodology have been applied above the requirements of TOOL 27 and therefore, the applicability condition has been met.

3.3 Project Boundary

To be completed in full validation version

3.4 Baseline Scenario

To be completed in full validation version

3.5 Additionality

To be completed in full validation version

3.5.1 Regulatory Surplus

To be completed in full validation version

3.5.2 Additionality Methods

To be completed in full validation version

3.6 Methodology Deviations

To be completed in full validation version

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

To be completed in full validation version

4.2 Project Emissions

To be completed in full validation version

4.3 Leakage Emissions

To be completed in full validation version

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

To be completed in full validation version

5 MONITORING

5.1 Data and Parameters Available at Validation

To be completed in full validation version

5.2 Data and Parameters Monitored

To be completed in full validation version

5.3 Monitoring Plan

To be completed in full validation version

6 APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

To be completed in full validation version

7 APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.